

Leonardo de Andrade Santos

Electrical Engineering | Digital IC, FPGA, DSP & Embedded Systems | Researcher in Digital Predistortion

Electronic engineer and MSc researcher working across FPGA-based digital design, embedded firmware, and software development, with hands-on experience in VHDL, C/C++, Python, and hardware-efficient Digital Predistortion architectures.

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EDUCATION

Master's Degree in Electrical Engineering

Federal University of Paraná (UFPR), Brazil

03/2025 - Present

Research

- Research area: digital circuits and signal processing for telecommunications.
- Thesis topic: hardware-efficient Digital Predistortion (DPD) architectures for RF power amplifier linearization.
- Developed and evaluated a delay-dependent Memory Polynomial model for power amplifier modeling, targeting reduced computational and hardware complexity.
- Implemented and verified the proposed architecture in VHDL, with FPGA-oriented validation and ASIC design flow exploration.
- Expected thesis defense: December 2026.
- Advisor: Prof. Eduardo Gonçalves de Lima.

Bachelor's Degree in Electrical Engineering

Federal University of Paraná (UFPR), Brazil

03/2019 - 12/2024

Focus

- Strong focus on electronics, digital systems, embedded systems, signal processing, and integrated circuits.
- Teaching assistant in Digital Electronics and Microelectronics.
- Participated in PET UFPR projects involving IoT, machine learning, 3D printing, technical courses, and student-led engineering initiatives.
- Member of UFPR Baja SAE electrical subsystem, working on wiring harnesses, PCBs, firmware, telemetry, and vehicle electronics.

PUBLICATIONS AND ACADEMIC OUTPUT

Power Amplifier Modeling Using Memory Polynomials with Delay-dependent Polynomial Orders (2026)

- Accepted for oral presentation at the 16th Workshop on Circuits and Systems Design (WCAS 2026).
- Work related to MSc research on hardware-efficient DPD models for RF power amplifier linearization.
- Proposed a Memory Polynomial variation with delay-dependent polynomial orders to balance modeling accuracy and hardware resource utilization.

MSc Dissertation — Digital Predistortion for RF Power Amplifiers (2025–2026)

- In progress. Research focuses on DPD modeling, fixed-point analysis, VHDL implementation, FPGA validation, and ASIC-oriented design exploration.

- Includes evaluation of Memory Polynomial models, variable-order structures, LUT-based implementations, and hardware cost versus modeling accuracy trade-offs.

FULL PAPERS IN CONFERENCE PROCEEDINGS

Digital predistorter described in VHDL and based on Memory Polynomial (2024)

Lima, E. G.; Franca, S. B. L.; Santos, L. A. — *SeMicro-PR, Curitiba, Brazil*

- Conference paper related to VHDL implementation of a Memory Polynomial-based digital predistorter.

Accuracy of Memory Polynomials with fixed-point processing (2023)

Lima, E. G.; Franca, S. B. L.; Santos, L. A. — *Paranaense Microelectronics Conference, Curitiba, Brazil*

- Conference paper focused on fixed-point processing accuracy in Memory Polynomial-based modeling.

Design of a dedicated integrated circuit for a TRNG based on ring oscillators (2022)

Santos, L. A.; Franca, S. B. L.; Lima, E. G. — *SeMicro-PR, Curitiba, Brazil*

- Conference paper on a dedicated integrated circuit design for a ring-oscillator-based True Random Number Generator.

ABSTRACTS IN CONFERENCE PROCEEDINGS

FPGA-based digital predistorters for wireless transmitters (2023)

Santos, L. A.; Lima, E. G.; Franca, S. B. L. — *14th SIEPE, Curitiba, Brazil*

Design of a dedicated integrated circuit for a TRNG based on ring oscillators (2022)

Santos, L. A.; Franca, S. B. L. — *13th SIEPE, Curitiba, Brazil*

SELECTED PRESENTATIONS AND ACADEMIC EVENTS

Paranaense Microelectronics Conference — Seminar presentation (2024)

Digital predistorter described in VHDL and based on Memory Polynomial

Paranaense Microelectronics Conference — Seminar presentation (2023)

Accuracy of Memory Polynomials with fixed-point processing

SIEPE — Seminar presentation (2023)

FPGA-based digital predistorters for wireless transmitters

Paranaense Microelectronics Conference — Seminar presentation (2022)

Design of a dedicated integrated circuit for a TRNG based on ring oscillators

SIEPE — Seminar presentation (2022)

Dedicated integrated circuit project for a TRNG based on ring oscillators

SIEPE — Seminar presentation (2021)

BigPET: a pedagogical and scientific experiment in data science and machine learning

SIEPE — Seminar presentation (2021)

PET 3D: encouraging maker culture within PET

RESEARCH EXPERIENCE

MSc Researcher — Digital Predistortion and Hardware-Efficient DSP

GICS-UFPR / Federal University of Paraná

03/2025 - Present

Research in circuits and systems for telecommunications

Research tasks

- Modeled RF power amplifier behavior using Memory Polynomial-based Digital Predistortion techniques.
- Evaluated floating-point and fixed-point implementations, including word-length optimization and integer-bound search.
- Developed a delay-dependent polynomial order approach, evaluating 125 model configurations and Pareto trade-offs between accuracy and complexity.
- Implemented the architecture in VHDL with pipelined datapaths and generate-based configurability.
- Verified numerical consistency between Python models and VHDL simulation results.
- Explored FPGA and ASIC implementation flows using Xilinx, simulation tools, and Cadence-based digital IC tools.

Undergraduate Researcher — TRNG Integrated Circuit

GICS-UFPR / Federal University of Paraná

08/2021 - 12/2024

Digital circuit design and integrated circuit implementation

Research tasks

- Designed and evaluated a True Random Number Generator (TRNG) based on ring oscillators.
- Implemented and validated the architecture in VHDL and FPGA-oriented flows.
- Designed the circuit using a 130 nm BiCMOS 8HP technology flow.
- Used Cadence tools including NCLaunch, Genus, and Innovus for simulation, synthesis, and physical design exploration.
- Validated random output quality using NIST statistical tests.

TEACHING AND ACADEMIC ACTIVITIES

Teaching Assistant — Digital Electronics and Microelectronics

Federal University of Paraná (UFPR)

2021 - 2024

Undergraduate teaching support

Teaching support

- Assisted students with digital electronics, microelectronics concepts, laboratory practices, and technical exercises.
- Supported practical activities involving digital logic, circuit analysis, and implementation-oriented topics.

Python Course Instructor

Federal University of Paraná (UFPR)

2021 - 2024

Technical training for Chemical Engineering students

Course topics

- Taught introductory Python programming for engineering applications.
- Covered programming fundamentals, problem solving, and practical computational examples.

Administrative Manager and Project Member

SEATEL / PET UFPR

08/2021 - 12/2024

Student academic and extension activities

Activities

- Organized courses, technical lectures, workshops, and academic activities.
- Participated in IoPET, BigPET, and PET3D projects involving IoT, machine learning, 3D printing, and educational outreach.

PROFESSIONAL EXPERIENCE

Embedded and Software Engineer

IoTag Tecnologia

10/2022 - 02/2026

Precision agriculture and embedded IoT systems

Achievements/Tasks

- Developed embedded firmware in C/C++ for ESP32/ESP8266-based devices and virtual terminal interfaces for agricultural machinery.
- Built Python backend services using Flask/FastAPI, SQLAlchemy, PostgreSQL, Cassandra, and AWS S3.
- Developed computer vision and data processing tools using OpenCV and Python.
- Worked with Linux-based development workflows, embedded debugging, cloud integration, and database-backed applications.
- Contributed to CI/CD and automation workflows for software deployment and maintenance.

Engineering Intern

Lactec

06/2022 - 10/2022

Technology and research institute

Achievements/Tasks

- Supported the development of capacitive water level and flow sensors for tidal energy generation systems.
- Worked on electronics, microcontroller programming, sensor integration, and experimental validation.

SELECTED TECHNICAL PROJECTS

Hardware-Efficient Digital Predistorter in VHDL (2025--2026)

- Implemented a configurable VHDL architecture for Memory Polynomial-based DPD.
- Used pipelined datapaths, fixed-point arithmetic, and delay-dependent polynomial orders.
- Compared FPGA-oriented resource utilization and modeling accuracy against conventional implementations.

TRNG Integrated Circuit in 130 nm Technology (2021--2024)

- Designed a ring-oscillator-based TRNG using VHDL and digital IC implementation tools.
- Synthesized and explored physical implementation using Cadence Genus and Innovus.
- Validated output randomness with NIST statistical tests.

UFPR Baja SAE Electrical System

(2019--2021)

- Developed wiring harnesses, telemetry support, embedded firmware, and PCB-related tasks for an off-road competition vehicle.
- Participated in system integration, electrical design, and vehicle testing.
- Team achievements included 1st place in the 2019 Regional South electrical project, 11th place nationally in 2020, and 2nd place in the 2020 Regional South electrical project.

IoPET Smart Lock (2021--2024)

- Developed an IoT electronic lock using ESP-01/ESP8266 and MQTT for remote control and embedded communication.

BigPET Fake News Detector (2021--2024)

- Led a machine learning project for text classification and fake news detection using Python.
- Explored models such as Logistic Regression, KNN, and Random Forest.

TECHNICAL SKILLS

VHDLVerilogFPGA designDigital IC design

ASIC flowFixed-point DSPDigital predistortion

RF power amplifier modelingPythonMATLABC/C++

Embedded LinuxESP32/ESP8266FastAPIFlask

PostgreSQLSQLAlchemyCassandraAWS S3

OpenCVGitCI/CD

EDA, SIMULATION AND DEVELOPMENT TOOLS

Cadence GenusCadence InnovusCadence NCLaunch

Xilinx ISEVivadoModelSim / QuestaGHDL

GTKWaveYosysADSAltium DesignerProteus

LinuxNeovim

CERTIFICATIONS AND COURSEWORK

Machine Learning with Python

60 hours

Databases and SQL

60 hours

Android Development

60 hours